

Locally robust commensurate Bayesian additive regression trees for historical information borrowing

Abstract

Historical control data can improve the precision of clinical-trial analyses, but residual differences between populations may vary with patient characteristics. We propose locally robust commensurate Bayesian additive regression trees, which represent historical prognosis and trial-specific discrepancy by separate sums of trees. A continuous spike-and-slab prior allows discrepancy contributions to shrink toward zero or depart from historical agreement. We define prior effective sample size for the current-control mean by matching uncertainty under a prior informed by real-world data (RWD) to that of a normal reference experiment and averaging the equivalent counts over a prior-scale distribution. This definition provides a reference for prior-scale selection and an explicit information ceiling. We derive marginal leaf likelihoods for posterior computation and, in fixed-region Gaussian benchmarks, establish learning of practical agreement and characterize estimation benefit and conflict risk. Simulations with Gaussian and censored survival outcomes illustrate precision gains under compatible controls and protection against bias from complete pooling, although source-adjusted BART remains competitive and small regional discrepancies are difficult to recover. An application to a single-arm multiple-myeloma trial reports restricted mean survival contrasts and historical information at the trial patients' covariate profiles. The implemented prior-scale selection and fitting distributions are distinguished from the theoretical prior effective sample size reference.

Keywords: Bayesian additive regression trees; commensurate prior; effective sample size; external controls; historical borrowing; survival analysis.

1 Introduction

Motivation. Historical control data can support estimation when a randomized control arm is small or a trial has a single treatment arm. Their contribution depends on how well historical outcomes represent control outcomes in the target population. In multiple myeloma, for example, an external registry may provide control information for a single-arm treatment study, but the cohorts can differ in patient characteristics and clinical management. Covariate adjustment addresses measured prognostic differences; residual source differences may remain and may vary across patient profiles.

Existing approaches. Power priors discount the historical likelihood, whereas commensurate priors link current and historical parameters through a hierarchical distribution (Ibrahim and Chen, 2000; Hobbs et al., 2011, 2012). Meta-analytic-predictive (MAP) priors summarize historical information for a new study, with robust extensions accommodating departures from historical predictions (Neuenschwander et al., 2010; Schmidli et al., 2014). Propensity-score-integrated methods use measured covariates to regulate the historical

contribution (Chen et al., 2020). These approaches address different aspects of comparability. A single study-level borrowing parameter does not explicitly represent how residual source differences vary across covariate profiles.

Modeling rationale. Bayesian additive regression trees (BART) approximate regression functions by regularized sums of trees (Chipman et al., 2010). BART-based external-data methods and covariate-adaptive borrowing designs extend adjustment beyond linear relationships (Zhou and Ji, 2021; Jin et al., 2023). Covariate adjustment and discrepancy modeling nevertheless have distinct roles: the former represents prognosis, whereas the latter describes residual differences between trial and historical controls at the same covariate values. Separately regularizing these functions allows a complex prognostic relationship to coexist with a simpler source discrepancy.

Contribution. We propose locally robust commensurate BART (lrcBART), with separate sums of trees for historical prognosis and trial-specific discrepancy. A continuous spike-and-slab prior on discrepancy terminal-node parameters provides local shrinkage toward historical agreement. We define prior effective sample size (ESS) for an RWD-informed prior on the current-control mean and use it as a reference for prior-scale selection. We also derive marginal leaf likelihoods for Bayesian backfitting. Joint regional Gaussian benchmarks establish learning of practical agreement and quantify the estimation benefit and conflict risk of borrowing; supporting leaf calculations describe the mixture mechanism. Gaussian and survival simulations assess operating characteristics, and a multiple-myeloma application illustrates standardized survival comparisons and patient-profile prior ESS. In single-arm studies, the discrepancy remains prior-driven because trial-control outcomes are unavailable.

2 Methods

2.1 Data structure and estimand

Data structure. Use group labels trt , 1 and 2 for randomized treatment, randomized control and RWD control, respectively. For $a \in \{\text{trt}, 1, 2\}$, let n_a denote the sample size, $Y_{a,i}$ the outcome and $x_{a,i}$ the covariate vector for participant $i = 1, \dots, n_a$, and σ_a^2 the residual variance. Define $\mathbf{Y}_a := (Y_{a,1}, \dots, Y_{a,n_a})$ and $\mathbf{X}_a := (x_{a,1}, \dots, x_{a,n_a})$. The trial size is $N := n_{\text{trt}} + n_1$. Standardization averages over the empirical covariate distribution of all trial participants. In a single-arm trial, $n_1 = 0$. For Gaussian outcomes, observations are conditionally independent given the functions and residual variances, with

$$\begin{aligned} Y_{\text{trt},i} \mid x_{\text{trt},i} &\sim \mathcal{N}\{f_{\text{trt}}(x_{\text{trt},i}), \sigma_{\text{trt}}^2\}, \\ Y_{1,i} \mid x_{1,i} &\sim \mathcal{N}\{f(x_{1,i}) + g(x_{1,i}), \sigma_1^2\}, \\ Y_{2,i} \mid x_{2,i} &\sim \mathcal{N}\{f(x_{2,i}), \sigma_2^2\}. \end{aligned} \tag{1}$$

Control discrepancy and estimand. The surface $g(x)$ is the conditional trial-minus-historical control difference. Both control sources inform f in the final joint analysis; only randomized-control residuals directly inform the discrepancy. The treatment surface f_{trt} is fitted separately. We standardize the treatment contrast to the empirical trial covariate distribution:

$$\Delta := \frac{1}{N} \sum_{a \in \{\text{trt}, 1\}} \sum_{i=1}^{n_a} \{f_{\text{trt}}(x_{a,i}) - f(x_{a,i}) - g(x_{a,i})\}. \tag{2}$$

Interpretation of Δ as a causal treatment effect requires consistency and appropriate treatment-assignment and transportability assumptions. Estimating trial-control outcomes from historical observations additionally requires adequate covariate support or explicit reliance on extrapolation.

2.2 Sum-of-trees representations and prior distributions

Tree representation. After subtracting a fixed outcome-centering constant, write

$$f(x) := \sum_{h^f=1}^{H_f} \sum_{\ell \in \mathcal{T}_{h^f}^f} \theta_{h^f \ell}^f \mathbf{1}\{x \in \ell\}, \quad g(x) := \sum_{h^g=1}^{H_g} \sum_{\ell \in \mathcal{T}_{h^g}^g} \theta_{h^g \ell}^g \mathbf{1}\{x \in \ell\}. \quad (3)$$

Here $\mathcal{T}_{h^f}^f$ and $\mathcal{T}_{h^g}^g$ denote prognostic and discrepancy tree structures, and $\theta_{h^f \ell}^f$ and $\theta_{h^g \ell}^g$ their terminal-node parameters. In each sum, ℓ indexes a terminal-node region. The two sums of trees have separate structures. Conditional on a prespecified admissible splitting rule, the tree prior uses a depth-dependent splitting probability. For $k \in \{f, g\}$,

$$P(\text{split at depth } d) = \alpha_k (1+d)^{-\beta_k}, \quad \theta_{h^k \ell}^k \mid \mathcal{T}_{h^k}^k \sim N(0, \lambda_{h^k}^2). \quad (4)$$

We use $(\alpha_f, \beta_f) = (0.95, 2)$ and $(\alpha_g, \beta_g) = (0.5, 3)$, encouraging a simpler discrepancy than outcome surface. Candidate splitting rules may be shared while tree structures are estimated separately.

Treatment model. The treatment function has the independent BART representation

$$f_{\text{trt}}(x) := \sum_{h^{\text{trt}}=1}^{H_{\text{trt}}} \sum_{\ell \in \mathcal{T}_{h^{\text{trt}}}^{\text{trt}}} \theta_{h^{\text{trt}} \ell}^{\text{trt}} \mathbf{1}\{x \in \ell\}.$$

We use $H_{\text{trt}} = 50$ trees, depth parameters $(\alpha_{\text{trt}}, \beta_{\text{trt}}) = (0.95, 2)$ and terminal-node variance $\lambda_{\text{trt}}^2 = R_{y, \text{trt}}^2 / (16H_{\text{trt}})$, where $R_{y, \text{trt}}$ is the range of the centered treated outcomes (observed log times for survival data). Its residual variance has the corresponding inverse-gamma prior described in Appendix A.1. The treatment fit is separate from the control model, including in application analyses with $H_f = 10$.

Discrepancy prior. For each discrepancy terminal node,

$$z_{h^g \ell} \mid w \sim \text{Bernoulli}(w),$$

$$\theta_{h^g \ell}^g \mid z_{h^g \ell}, \tau_0^2, \tau_1^2 \sim \begin{cases} N(0, \tau_0^2), & z_{h^g \ell} = 1, \\ N(0, \tau_1^2), & z_{h^g \ell} = 0, \end{cases} \quad 0 < \tau_0^2 < \tau_1^2. \quad (5)$$

Terminal-node indicators are independent conditional on w and the trees; their parameters are independent conditional on the indicators, variances and trees. The continuous spike shrinks node contributions toward zero, whereas the slab permits larger discrepancies of either sign. The latent $z_{h^g \ell}$ is distinct from observed source membership. The probability w is shared across discrepancy nodes.

Hyperparameter specification. The default two-arm hierarchical specification is

$$w \sim \text{Beta}(1, 1), \quad \tau_0^2 \sim \text{scaled-Inv } \chi^2(v_0, s_0^2) \text{ restricted to } (0, \tau_1^2), \quad v_0 = 3. \quad (6)$$

The slab variance is fixed, and residual variances have inverse-gamma priors. Appendix A gives their scale specifications and conditional distributions. Prior ESS calibration guides selection of s_0^2 . Two-arm fitting then updates w , τ_0^2 , tree structures and terminal-node parameters. The single-arm analyses use fixed w and $\tau_0^2 = \min(s_0^2, \tau_1^2/2)$. Section 2.3 distinguishes the prior ESS reference distribution from these fitting distributions.

2.3 Prior effective sample size

ESS principle. Prior ESS expresses the information supplied by a prior in units of observations from a reference likelihood. This requires an information measure for the prior and the information supplied by one reference observation. The expected local-information-ratio (ELIR) approach illustrates this principle by comparing local log-prior curvature with one-observation Fisher information and averaging the resulting ratios over the prior (Neuenschwander et al., 2020). We use the same information-matching principle, with inverse conditional marginal variance as the information measure for the prior under study.

Current-control mean and prior construction. Let μ denote the current-control mean, obtained by averaging the current-control response surface $f + g$ over the empirical covariate distribution of the randomized control arm. A preliminary fit of f uses $(\mathbf{Y}_2, \mathbf{X}_2)$ and is evaluated at $\mathbf{X}_1$; an independent prior for g represents the current-minus-historical discrepancy. Define

$$\begin{aligned} \mu_f &:= n_1^{-1} \sum_{i=1}^{n_1} f(x_{1,i}), & \mu_g &:= n_1^{-1} \sum_{i=1}^{n_1} g(x_{1,i}), & \mu &:= \mu_f + \mu_g, \\ V_f &:= \text{Var}(\mu_f \mid \mathbf{Y}_2, \mathbf{X}_2, \mathbf{X}_1), & V_g(w, \tau_0^2) &:= \text{Var}(\mu_g \mid \mathbf{X}_1, w, \tau_0^2). \end{aligned} \quad (7)$$

The posterior distribution of μ_f from the historical fit and the prior distribution of μ_g induce the RWD-informed prior $\pi(\mu \mid \mathbf{Y}_2, \mathbf{X}_2, \mathbf{X}_1, w, \tau_0^2)$. Conditional independence of the historical posterior for f and the prior for g gives

$$V_\pi(w, \tau_0^2) := \text{Var}_\pi(\mu \mid \mathbf{Y}_2, \mathbf{X}_2, \mathbf{X}_1, w, \tau_0^2) = V_f + V_g(w, \tau_0^2). \quad (8)$$

Here V_f retains historical-tree uncertainty and dependence among predictions at the randomized-control profiles. For discrepancy-tree collection $\mathcal{T}^g := (\mathcal{T}_{h^s}^g)_{h^s=1}^{H_g}$, let

$$a_{h^s \ell} := n_1^{-1} \sum_{i=1}^{n_1} \mathbf{1}\{x_{1,i} \in \ell \text{ of tree } h^s\}.$$

Integrating the zero-mean leaf parameters, indicators and prior trees yields

$$V_g(w, \tau_0^2) = \{w\tau_0^2 + (1-w)\tau_1^2\} \mathbb{E}_{\mathcal{T}^g} \left[\sum_{h^s, \ell} a_{h^s \ell}^2 \right]. \quad (9)$$

Information matching. We measure the information in the RWD-informed prior by inverse conditional marginal variance,

$$\mathcal{J}_\pi(w, \tau_0^2) := \frac{1}{V_\pi(w, \tau_0^2)} = \{V_f + V_g(w, \tau_0^2)\}^{-1}.$$

One observation from the normal current-control reference likelihood $Y \mid \mu \sim \mathcal{N}(\mu, \sigma_1^2)$ supplies expected

Fisher information $\mathcal{J}_{\text{ref}}(\mu) := i_F(\mu) = 1/\sigma_1^2$. For fixed w and τ_0^2 , define the equivalent count by matching the information in the RWD-informed prior to the information in $m(w, \tau_0^2)$ reference observations:

$$\mathcal{J}_\pi(w, \tau_0^2) = m(w, \tau_0^2) \mathcal{J}_{\text{ref}}(\mu), \quad m(w, \tau_0^2) = \frac{\sigma_1^2}{V_f + V_g(w, \tau_0^2)}. \quad (10)$$

The prior ESS averages these conditional counts over the spike-variance calibration distribution indexed by s_0^2 :

$$\text{ESS}(w, s_0^2) := \mathbb{E}_{\tau_0^2} \{m(w, \tau_0^2)\} = \mathbb{E}_{\tau_0^2} \left[\frac{\sigma_1^2}{V_f + V_g(w, \tau_0^2)} \right]. \quad (11)$$

Appendix B.1 compares this definition with other prior-ESS methods.

Calibration and interpretation. Prior-scale calibration sets $w = 1$ and uses the *untruncated* scaled inverse-chi-square law with parameters (v_0, s_0^2) :

$$V_g(1, \tau_0^2) = \tau_0^2 \mathbb{E}_{\mathcal{T}^s} \left[\sum_{h^s, \ell} a_{h^s, \ell}^2 \right], \quad (12)$$

$$\text{ESS}_{\tau_0}(s_0^2) := \text{ESS}(1, s_0^2) = \mathbb{E}_{\tau_0^2} \left[\frac{\sigma_1^2}{V_f + V_g(1, \tau_0^2)} \right].$$

Theorem 1 (Prior ESS curve). *Fix $0 < V_f < \infty$, $0 < \sigma_1^2 < \infty$, $v_0 > 0$ and a discrepancy-tree prior with $H_g \geq 1$ and finitely many terminal nodes almost surely. Under the untruncated calibration law, $\text{ESS}_{\tau_0}(s_0^2)$ is continuous, strictly decreasing and strictly convex on $s_0^2 > 0$. Its limits are σ_1^2/V_f as $s_0^2 \downarrow 0$ and zero as $s_0^2 \rightarrow \infty$. Each target strictly between those limits has a unique positive solution.*

Consequently, the upper limit is

$$\text{ESS}_{\text{ceiling}} := \lim_{s_0^2 \downarrow 0} \text{ESS}_{\tau_0}(s_0^2) = \frac{\sigma_1^2}{V_f}. \quad (13)$$

Appendix B.2–B.4 derives the variance terms, proves the theorem and describes Monte Carlo evaluation. The theoretical reference holds σ_1^2 fixed. Numerical calibration substitutes a fixed estimate $\hat{\sigma}_1^2$ rather than averaging over final-model posterior draws; in two-arm analyses this estimate uses trial controls. The reported block-based prior-scale selector and the differences between the calibration and fitting priors are described in Appendix B.5 and B.7. Tables S9 and S11(a) distinguish target, block-based and full-variance summaries.

An ESS of m means that the RWD-informed prior supplies, on average over the calibration distribution of τ_0^2 , the same conditional precision about the specified current-control mean as m independent normal-reference observations. Patient-profile ESS values do not generally sum or average to overall ESS because predictions at different profiles are correlated and inverse variance is nonlinear.

2.4 Posterior computation

We use Bayesian backfitting, updating one tree conditional on all other trees and shared parameters. An f -tree update uses partial residuals from both control sources with their respective residual precisions. A g -tree update uses trial-control partial residuals only. The terminal-node parameter is integrated out when proposing

a tree structure; for g , the latent mixture indicator is also integrated out. Metropolis–Hastings updates therefore use products of closed-form marginal leaf factors. After retaining a tree, its node indicators and parameters are sampled from their conditional distributions. Shared variances and the mixture probability are updated separately. Appendix A gives the derivation, the acceptance probability and the sampling algorithm.

2.5 Local agreement and regional borrowing properties

Local discrepancy summary. For a prespecified tolerance $c > 0$, define

$$B_c(x) := P\{|g(x)| < c \mid \text{observed data}\}. \quad (14)$$

This is the posterior probability of practical agreement between the control sources at covariate profile x . Its interpretation depends on the outcome scale and the chosen tolerance. It is neither a likelihood weight nor prior ESS.

Joint regional model. We study two properties: learning practical agreement and characterizing the estimation benefit and error induced by borrowing. The following results use a Gaussian model on a fixed finite partition $R_1, \dots, R_J$ of the target covariate domain. Within region j , independent historical and trial controls have means f_j and $f_j + g_j$, with known positive residual variances σ_2^2 and σ_1^2 . Here g_j is the total regional source difference. Both source means are unknown and estimated jointly. Use independent priors across regions, with $f_j \sim N(0, \lambda_j^2)$ independent of

$$g_j \sim wN(0, \tau_0^2) + (1-w)N(0, \tau_1^2), \quad 0 < w < 1, \quad 0 < \tau_0^2 < \tau_1^2 < \infty.$$

All prior parameters are fixed, with $0 < \lambda_j^2 < \infty$.

Theorem 2 (Learning local practical agreement). *Suppose the regional model is correctly specified, with true parameters f_j^*, g_j^* . As the numbers $n_{1,j}$ and $n_{2,j}$ of controls from both sources tend to infinity in every region, for every $\varepsilon > 0$,*

$$\Pi \left\{ \max_{1 \leq j \leq J} (|f_j - f_j^*| + |g_j - g_j^*|) > \varepsilon \mid \mathcal{D} \right\} \rightarrow 0$$

almost surely under the true sampling law, where $\mathcal{D}$ denotes the observed control data. Thus the current-control means $f_j + g_j$ are also consistently learned. If $|g_j^| \neq c$ in every region, the following limits hold simultaneously:*

$$B_{c,j} := \Pi(|g_j| < c \mid \mathcal{D}) \rightarrow \begin{cases} 1, & |g_j^*| < c, \\ 0, & |g_j^*| > c, \end{cases} \quad \text{almost surely.} \quad (15)$$

For $x \in R_j$, $B_c(x) = B_{c,j}$. The posterior increasingly supports practical agreement where the true difference is small and rules it out where the difference exceeds the tolerance. This interpretation uses posterior probabilities directly, without a labeling rule.

Estimation benefit and conflict risk. For an exact risk calculation in one region, retain the discrepancy mixture but replace the normal prior on the historical mean f by the flat prior $p(f) \propto 1$. This is a separate benchmark assumption. Write $\mu_1 = f + g$, $d = \mu_1 - f$ for the true discrepancy, $v_s = \sigma_s^2/n_s > 0$, $V = v_1 + v_2$,

and $D = \bar{Y}_1 - \bar{Y}_2$. Let

$$\begin{aligned}\gamma(D) &= \frac{w\phi(D; 0, V + \tau_0^2)}{w\phi(D; 0, V + \tau_0^2) + (1-w)\phi(D; 0, V + \tau_1^2)}, \\ a_k &= \frac{v_1}{V + \tau_k^2}, \quad a(D) = \gamma(D)a_0 + \{1 - \gamma(D)\}a_1,\end{aligned}\tag{16}$$

where $\phi(\cdot; m, v)$ is a normal density with mean m and variance v .

Theorem 3 (Regional borrowing benefit and conflict risk). *Under the flat-baseline regional benchmark, the posterior mean of the current-control mean is*

$$\hat{\mu}_1 = \{1 - a(D)\}\bar{Y}_1 + a(D)\bar{Y}_2.\tag{17}$$

The coefficient $a(D)$ decreases strictly with $|D| > 0$ and tends to $a_1 > 0$ as $|D| \rightarrow \infty$. Put $a_* = v_1/V$. Its repeated-sampling bias and mean squared error are

$$\begin{aligned}\text{Bias}_d(\hat{\mu}_1) &= -\text{E}_d\{a(D)D\}, \\ R(d) &:= \text{E}_d(\hat{\mu}_1 - \mu_1)^2 = \frac{v_1 v_2}{V} + \text{E}_d[\{(a_* - a(D))D - a_* d\}^2],\end{aligned}\tag{18}$$

where the expectations on the right are over $D \sim N(d, V)$. Under exact agreement, $d = 0$, the estimator is unbiased and $R(0) < v_1$, the mean squared error of the trial-only mean. For fixed sample sizes and prior parameters, as $|d| \rightarrow \infty$,

$$\frac{\text{Bias}_d(\hat{\mu}_1)}{d} \longrightarrow -a_1, \quad \frac{R(d)}{d^2} \longrightarrow a_1^2.$$

Theorem 3 establishes an estimation benefit under agreement and quantifies the remaining error under conflict. The finite normal slab leaves residual shrinkage, so the benchmark does not have uniformly bounded borrowing bias over arbitrarily large discrepancies. The mean squared error result does not imply that every realized posterior interval is shorter.

Appendix C gives the proofs, implications for standardized contrasts, and supporting fixed-leaf calculations. These regional results account for uncertainty in both source means, but do not establish consistency or risk guarantees for learned tree ensembles, updated shared hyperparameters, data-dependent prior scales, or censored outcomes. The flat-baseline risk calculation is not a result for the proper Gaussian baseline prior in Theorem 2. Learning the total discrepancy under the full model remains a theoretical objective.

2.6 Survival outcomes and single-arm trials

Survival outcomes. For survival outcomes, apply (1) to latent log event times. Event and right-censoring contributions enter the likelihood, and censored log times are imputed from lower-truncated normal distributions. At each posterior draw, the control survival curve and its trial-standardized version are

$$S_1(t | x) = 1 - \Phi\left(\frac{\log t - f(x) - g(x)}{\sigma_1}\right), \quad \bar{S}_1(t) := \frac{1}{N} \sum_{a \in \{\text{trt}, 1\}} \sum_{i=1}^{n_a} S_1(t | x_{a,i}).\tag{19}$$

The treated curve is defined using $f_{\text{trt}}, \sigma_{\text{trt}}$. We derive population medians and restricted mean survival times (RMSTs) from these standardized curves, rather than averaging individual medians. Appendix D describes

censoring, survival summaries and numerical conventions. Survival ESS uses an uncensored normal reference on the log-time scale; it has not been calibrated to a trial with the same censoring pattern.

Single-arm trials. The model also applies when all trial participants receive treatment. The target control surface remains $f + g$, with f informed by historical outcomes and g governed entirely by its prior. Treated outcomes inform the separate f_{tr} surface and do not identify the control discrepancy. Borrowing in a single-arm analysis therefore depends on the specified discrepancy prior and the historical-to-trial variance assumption.

3 Simulation studies

3.1 Design

Study designs. We examined four main study designs with 100 independently generated datasets per scenario setting (Table 1). Two-arm studies included 300 trial participants and 300 historical controls, with treatment probabilities $2/3$ for Gaussian outcomes and $3/4$ for survival outcomes. The main single-arm designs included 200 treated participants and 300 historical controls. A supplementary survival single-arm experiment used 30 treated participants. Ten predictors generated a nonlinear outcome surface; the historical cohort was regenerated for every replicate.

Data-generating scenarios. The two-arm studies considered common covariate distributions (Sc1), selection on measured prognostic variables (Sc2), selection on latent prognostic variables observed through imperfect proxies (Sc3), a global historical outcome shift (Sc4), and a regional historical shift (Sc5). Sc3 used correlations $-0.5, 0, 0.5$; Sc4 used shifts 1 and 2; Sc5 varied the shifted region and magnitude. These give 12 settings. Figure 1 depicts the source-selection mechanisms in Sc1–Sc3. The single-arm studies considered Sc1 and Sc2. Gaussian treatment effects were 1; survival effects were $\log(1.4)$ on the log-time scale. Appendix E gives the complete data-generating mechanism and the design-specific constants.

3.2 Methods and performance measures

Comparators. Two-arm comparators included linear models (LM), accelerated failure time (AFT) models and BART, with no historical pooling (-NP), complete pooling (-CP), or source-adjusted pooling (-PP), together with hierarchical regression. The -PP variants include a source indicator; hierarchical regression instead pools source-specific coefficients through their prior. Gaussian studies additionally included MAP and propensity-score-integrated composite likelihood (PSCL). Primary lrcBART analyses used $H_f = 50$, $H_g = 5$ and requested prior ESS 100. Two-arm analyses estimated w ; primary single-arm analyses fixed $w = 1$. Single-arm comparators included historical-control regression and hierarchical regression with discrepancy-variance prior scale $s_\tau^2 = 0.05$, as defined in Appendix E.4.

Performance measures. We report bias, root mean squared error (RMSE), 95% interval coverage and empirical power under prespecified decision rules. Bayesian benefit decisions require posterior probability greater than 0.95 that the mean difference exceeds 0.5 (Gaussian outcomes) or the survival ratio exceeds 1. PSCL requires its lower confidence limit to exceed the Gaussian threshold. Type I error was not evaluated, so the decision rules are not compared at a common demonstrated Type I error rate. Supplementary Tables S5–S8 additionally report interval width, posterior SD or standard error as appropriate, and Monte Carlo standard

errors. With 100 replicates, a proportion has Monte Carlo standard error at most 0.05. The primary survival estimand is the ratio of population median survival times. Supplementary RMST-ratio summaries use three-year and five-year horizons for the two-arm and single-arm studies, respectively, with the modified numerical functionals defined in Appendix D.5.

3.3 Gaussian two-arm results

Compatible and covariate-shift settings. Under Sc1, lrcBART had RMSE 0.174, compared with 0.209 for trial-only BART and 0.153 for pooled BART (Table 2). Coverage was 0.95, 0.97 and 0.95, respectively. Thus the discrepancy prior retained some of the precision available from pooling while allowing uncertainty about source agreement. Under Sc2, lrcBART had negligible mean bias (0.003), RMSE 0.186 and coverage 0.94; the corresponding pooled linear model had bias 0.343 and coverage 0.73.

Global and regional discrepancies. Under the global shift of 1, pooled BART had bias -0.689 and zero coverage, whereas lrcBART had bias -0.007 , RMSE 0.201 and coverage 0.94. With an uninformative proxy in Sc3 ($\rho = 0$), lrcBART and source-adjusted BART had RMSEs 0.315 and 0.309; pooled BART had bias 0.957. Under the regional X_5 shift of 1, lrcBART had RMSE 0.214, compared with 0.195 for source-adjusted BART and 0.209 for trial-only BART. The method therefore protected against severe pooling bias without uniformly dominating these alternatives.

3.4 Survival two-arm results

The survival results showed a similar precision–robustness trade-off (Table 3). Under Sc1, median-ratio RMSE was 0.290 for lrcBART, 0.392 for trial-only BART and 0.218 for pooled BART. Under a global historical log-time shift of 1, lrcBART had bias -0.007 and coverage 0.94, whereas pooled BART had bias -0.729 and zero coverage. For Sc3 with $\rho = 0$, lrcBART had RMSE 0.472 and coverage 0.97, compared with 2.802 and zero for pooled BART. Source-adjusted BART remained competitive, with RMSE 0.446 in that setting. At the regional X_5 shift of 1, lrcBART and source-adjusted BART had RMSEs 0.331 and 0.316.

3.5 Single-arm results

Gaussian outcomes. For Gaussian outcomes, primary lrcBART and BART-CP had similar RMSEs: 0.156 versus 0.153 in Sc1 and 0.177 versus 0.176 in Sc2 (Table 4). The additional discrepancy uncertainty increased coverage to 0.98 in Sc2, with an empirical power of 0.91. The retained hierarchical regression comparator had complete empirical coverage, with power 0.09 in Sc1 and 0.50 in Sc2.

Survival outcomes. For survival outcomes, primary lrcBART median-ratio RMSE was 0.178 in Sc1 and 0.216 in Sc2, with empirical power 0.59 and 0.73 (Table 4). Results for the smaller single-arm survival trial are reported in Supplementary Table S8. These results do not demonstrate data-adaptive rejection of an incompatible historical control surface in a single-arm trial.

3.6 Regional behavior and sensitivity analyses

Regional adaptation. Figure 2 displays the Gaussian probability of a small discrepancy and estimated discrepancy against X_5 for regional shifts of 0.5, 1 and 2 in Sc5, using the first 40 replicates. Across all

100 replicates, regional summaries showed the following pattern. At a shift of 0.5, mean small-discrepancy probabilities were similar inside and outside the compatible region. At a shift of 2, the Gaussian probabilities were 0.651 inside and 0.058 outside; the survival probabilities were 0.473 and 0.159. The mean discrepancy remained biased toward zero in the shifted region, particularly for survival outcomes. These results support a qualified claim of regional adaptation, with limited separation for small shifts and incomplete recovery even when shifts are larger.

Sensitivity analyses. Tables S4–S8 report all evaluated configurations, including two-arm prior ESS targets 75 and 50 and fractions 0.90, 0.50 and 0.25 of the information ceiling. Single-arm sensitivities include fixed $w = 0.9$, a target of 90% of the ceiling and the minimum-scale setting $s_0^2 = 10^{-6}$. The latter approximates pooling while retaining positive discrepancy variance. Requested targets need not equal the numerical ESS at the selected scale: for Gaussian Sc1, target 100 gave mean block-based ESS 82.5, and 21% of requests were capped. Figure S1 illustrates the lrcBART and MAP prior ESS curves for one Gaussian two-arm replicate.

4 Multiple-myeloma application

4.1 Cohorts and covariate harmonization

Study populations. The analysis extract contains 30 EloKRd trial patients and 200 triplet-treated historical controls from the University of Chicago Multiple Myeloma (UCMM) registry. EloKRd combines elotuzumab, carfilzomib, lenalidomide and dexamethasone; a published phase 2 study of this combination provides clinical context (Derman et al., 2022). The analysis uses the 30-patient extract available for this comparison. Table 5 summarizes age, sex, race, Hispanic/Latino status, cytogenetic risk and autologous stem-cell transplantation (ASCT). Race is represented by Black and Other indicators with White as the reference, yielding seven numeric predictors across six dimensions. Cohort selection is shown in Appendix Figure S2; endpoint construction and coding rules appear in Appendix F and Table S10.

Covariate imbalance. The cohorts differ markedly: high-risk cytogenetics occurred in 50.0% of EloKRd patients and 25.5% of UCMM controls, and ASCT in 30.0% and 87.0%. ASCT is a postinduction characteristic. Adjustment for it does not define a conventional baseline-standardized causal treatment effect; the reported comparisons are descriptive, conditional model-based contrasts. Residual confounding, selection and differences in clinical management remain possible.

4.2 Estimands and analysis settings

Endpoints and estimands. We analyze progression-free survival (PFS) and overall survival (OS), with times in years. For arm $a \in \{\text{trt}, 1\}$, five-year RMST and its contrasts are

$$R_a(5) = \int_0^5 \bar{S}_a(t) dt, \quad \Delta_R := R_{\text{trt}}(5) - R_1(5), \quad Q_R := \frac{R_{\text{trt}}(5)}{R_1(5)}. \quad (20)$$

Analysis specification. Adjusted methods standardize to the EloKRd covariates. Primary lrcBART uses $H_f = 10$, $H_g = 5$, fixed $w = 1$ and requested ESS 30. Each of four chains contributes 2,000 retained draws after 2,000 warm-up iterations. Comparators are unadjusted Kaplan–Meier (KM), a fully weighted historical AFT model (AFT-CP), BART-CP and hierarchical AFT. Main Table 6 reports the hierarchical discrepancy-

variance prior scale $s_\tau^2 = 0.05$; the $s_\tau^2 = 0.5$ sensitivity appears in Supplementary Table S11. The reference variance for single-arm ESS is estimated from UCMM under the equal-residual-variance assumption.

4.3 Survival comparisons

Progression-free survival. For PFS, primary lrcBART estimated five-year RMSTs of 3.521 years for EloKRd treatment and 3.535 years for its hypothetical control (Table 6). The posterior median ratio was 0.997 (95% credible interval 0.799–1.229), and the difference was -0.009 years (-0.768 to 0.714). BART-CP gave a similar ratio, 0.988 (0.796–1.215). The unadjusted KM ratio was 1.040, whereas AFT-CP gave 0.901, indicating sensitivity to outcome modeling and adjustment.

Overall survival. For OS, the primary lrcBART ratio was 0.898 (0.765–1.024), with a difference of -0.453 years (-1.065 to 0.101). The lrcBART intervals included a ratio of 1 and a difference of zero for both endpoints. AFT-CP yielded an OS ratio of 0.809 (0.654–0.951), whereas hierarchical AFT with $s_\tau^2 = 0.05$ gave 0.961 (0.761–1.387), indicating sensitivity to the control-model specification.

4.4 Historical information at EloKRd PFS profiles

Patient-profile targets. Table 7 reports prior ESS for the conditional mean control log-PFS at each of the 30 EloKRd covariate profiles. Rows are arranged in age groups (< 50 , $50-59$, $60-69$ and ≥ 70 years). The UCMM count in each row is the number of triplet-treated historical controls matching that row’s age group, sex, race, Hispanic/Latino status, cytogenetic risk and ASCT. The age category is a display summary; exact age remains in each profile x_i . For patient i , define

$$\begin{aligned}\mu_i &:= f(x_i) + g(x_i), \\ \text{ESS}_{\tau_0}(s_0^2; x_i) &:= \text{E}_{\tau_0^2} \left[\frac{\sigma_1^2}{V_f(x_i) + H_g \tau_0^2} \right], \\ V_f(x_i) &:= \text{Var}\{f(x_i) \mid \mathbf{Y}_2, \mathbf{X}_2, x_i\}.\end{aligned}\tag{21}$$

At each profile, the target and reference describe the same scalar parameter: the reference experiment is $Y_{ij}^{\text{ref}} \mid x_i \sim N(\mu_i, \sigma_1^2)$. The preliminary $H_f = 10$ historical model supplies $V_f(x_i)$, while the all-spike discrepancy reference supplies $H_g \tau_0^2$. Every row uses the full 200-patient triplet UCMM fit, the same residual variance, and the globally selected PFS scale $s_0^2 = 0.00398107$.

Information interpretation. Patient-profile prior ESS ranged from 3.8 to 24.5. An ESS of 20 means that the UCMM-informed prior supplies, on average over the scale law, the same conditional precision about μ_i as approximately 20 uncensored normal-reference controls with covariate profile x_i . These values describe prior information on the log-PFS scale. They are not equivalent counts under the trial’s censoring distribution and do not measure compatibility with unobserved EloKRd controls. The 30 pointwise values neither sum nor generally average to the overall ESS because the overall target depends on covariance among profile predictions and a nonlinear variance-to-ESS transformation (Appendix B.6).

4.5 Sensitivity and computation

Across the 24 lrcBART settings per endpoint, PFS ratio medians ranged from 0.955 to 0.964 and OS ratio medians from 0.885 to 0.892 (Table S11). These analyses vary H_f , fixed w and prior ESS targets. The primary block-based ESS was 26.3 for PFS and 25.0 for OS; the full-variance information ceilings were 43.1 and 24.1. Table S11(a) compares block-based and full-variance prior ESS. The largest $\hat{R}$ for the log RMST ratio was 1.0007 (Table S11(b)); these scalar diagnostics do not assess every aspect of tree-space exploration.

5 Discussion

Main findings. Separate sums of trees for historical prognosis and trial discrepancy allow their complexity and shrinkage to be controlled independently. In the two-arm simulations, this structure reduced bias from incompatible historical controls while retaining precision gains under compatible controls. Source-adjusted BART remained competitive, and small regional discrepancies were difficult to recover. The results therefore support evaluating both estimation accuracy and local discrepancy summaries when assessing historical borrowing.

Prior-information interpretation. Prior ESS provides an estimand-specific description of historical information. It matches conditional uncertainty about a current-control mean to a normal reference experiment and averages the equivalent counts over a specified scale distribution. Its ceiling depends on uncertainty in the historical fit for that mean. Applying the same definition at each trial covariate profile yields interpretable local information summaries without partitioning historical patients into independent borrowing contributions.

Limitations. Several limitations merit attention. First, the block-based selection procedure, spike-variance distribution and tree support used in the reported fits differ from the theoretical prior ESS reference. Aligning these components would permit prior ESS to describe the fitting prior directly; the present results instead use it as a prior-scale selection reference. Second, single-arm trials provide no outcome information about the trial-control discrepancy under the separate treatment model. Their results depend on the discrepancy prior, covariate support and residual-variance assumptions. The application also adjusts for a postinduction characteristic and is interpreted as a standardized model-based comparison rather than an identified causal effect.

Third, survival prior ESS is expressed in uncensored normal-reference units on the log-time scale. A reference design incorporating censoring would be required to translate it into an equivalent trial sample size. The supplementary simulation RMST summaries also use the modified functionals specified in Appendix D.5. Finally, 100 replicates provide limited precision for operating-characteristic comparisons, and Type I error was not evaluated. The regional theoretical results require fixed partitions and prior parameters; their learning and risk conclusions have not been established for the fitted tree ensembles. In particular, the risk calculation uses a flat baseline prior, and the finite normal slab retains shrinkage under extreme conflict.

Further work. Further work could align the prior ESS and fitting specifications, evaluate decision rules under null scenarios, and investigate prior sensitivity and discrepancy recovery over a broader range of source differences. A reference survival experiment with prespecified censoring would extend the information interpretation to trial-design settings.

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